

Eclipse IDE Cheat Sheet

Beginner's guide on navigating through Eclipse



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Eclipse IDE

Different IDEs can be used for different languages and use cases. [10 Best Java IDE For Developers in 2025 - GeeksforGeeks](#)

Why and IDE?

- Makes coding easier and faster
- Helps you avoid mistakes
- Organizes your project files in one place
- Gives you immediate feedback while you work

Why Eclipse?

- Free & Reliable — No cost, works across Windows, macOS, and Linux.
- Industry-Standard Tool — Still widely used in Java development.
- Java-Focused Features — Offers auto-completion, auto-formatting, debugging, and more to make coding smoother.
- Real-World Learning — Helps you practice organization, testing, and debugging in a professional environment.

First set up [Git and Github](#) and then [Setup Eclipse](#) on Windows or Mac OS



1. On the top menu, select **File->New-> Java Project** to display the New project dialog

Create Project JDK-25

The screenshot shows the 'New Java Project' dialog box in the Eclipse IDE. The dialog is titled 'New Java Project' and contains several sections for configuring a new project. Annotations with arrows point to specific fields and options:

- Project name:** The text 'ProjectName' is entered in the field. An annotation says: 'Give the project a name **FirstProject**. Do not use any spaces'.
- Location:** The path '/Users/debteach/Documents/GitHubRepos/CS1050_Fall2024/EclipseWorkSpace/ProjectName' is entered. An annotation says: 'This should already be set to the EclipseWorkSpace in your GitHubRepos/CSJavaRepos folder'.
- JRE:** Under 'Use an execution environment JRE:', 'JavaSE-24' is selected. An annotation says: 'This should be the latest version you installed JavaSE25 or some might need to select Default depending on how you set it up.'.
- Project layout:** Under 'Use project folder as root for sources and class files', the radio button is selected. An annotation says: 'Select use project folder'.
- Working sets:** The 'Add project to working sets' checkbox is unchecked.
- Module:** The 'Create module-info.java file' checkbox is checked. The 'Module name' field is empty. The 'Generate comments' checkbox is also checked.

At the bottom of the dialog, there are buttons for '< Back', 'Next >', 'Cancel', and 'Finish'.

This should already be set to the EclipseWorkSpace in your GitHubRepos/CSJavaRepos folder

This should be the latest version you installed JavaSE25 or some might need to select Default depending on how you set it up.

Select use project folder

Make sure all settings are correct. Click Finish.

Create Class & Java File

1. Select the project folder you want to create it in - in the package explorer on the left, click on your project to highlight it
2. Create a new class - click file > new > class
3. Configure settings -
 - a. Source folder should be set, if not, select the project folder you want it to go in
 - b. Package should be empty, if not, delete the package
 - c. Enter a name for your class (e.g., GEM01Calculation)
 - d. Modifiers: checkmark "public"
 - e. Method stubs: checkmark "public static void main(String[] args)"
 - f. Comments: checkmark "generate comments"
4. Click finish to create the class
5. This create your java file with class and main method

Step 1

Step 2

Step 3a

Step 3b

Step 3c

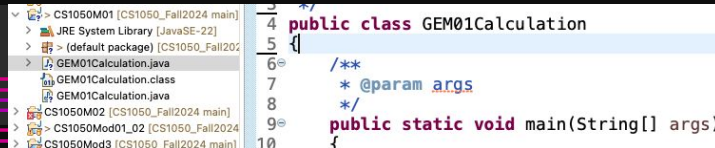
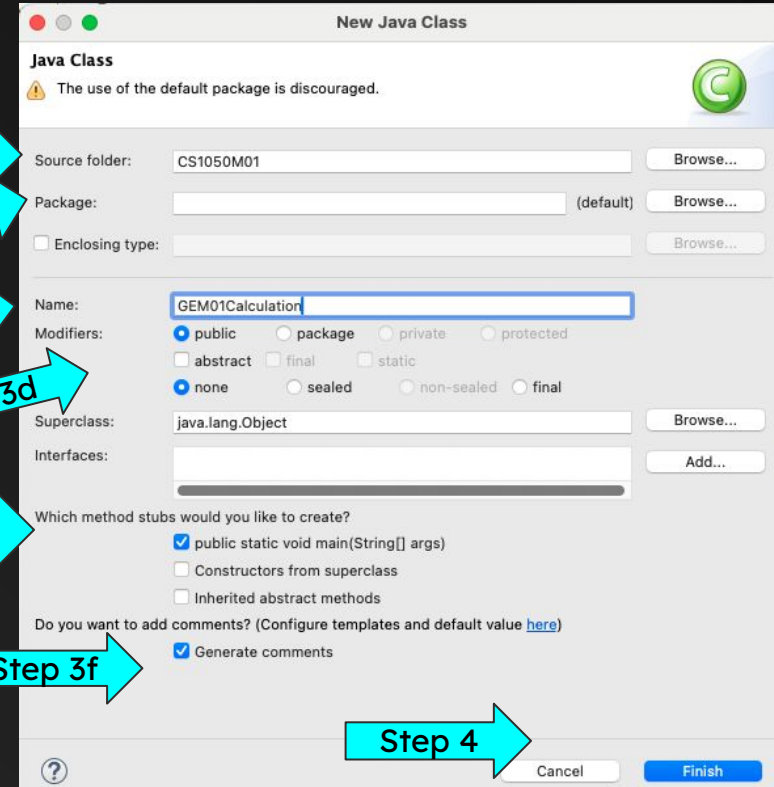
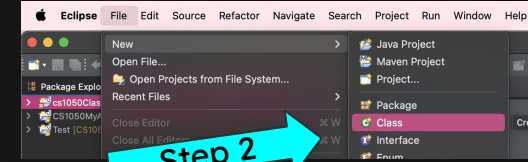
Step 3d

Step 3e

Step 3f

Step 5

Step 4



Navigating Java Perspective

Running your code

The screenshot shows the Eclipse IDE interface with the following components and annotations:

- Menus:** A red arrow points to the top menu bar (File, Edit, etc.).
- Run Program:** A red arrow points to the Run button (a green play icon) in the toolbar.
- Toolbar:** A red arrow points to the toolbar area on the right side of the IDE.
- Project Explore:** A red arrow points to the Project Explorer on the left side of the IDE.
- Project Folders:** A red arrow points to the project folders listed in the Project Explorer.
- Project contain JRE System and java files:** A red arrow points to the JRE System Library and java files listed in the Project Explorer.
- Edit code:** A red arrow points to the code editor area in the center of the IDE.
- Console Tab:** A red arrow points to the Console tab at the bottom of the IDE.
- Console display:** A red arrow points to the text "Hello Java" displayed in the Console.

The code in the editor is as follows:

```
1 /**
2  *
3  */
4 public class GEM01Calculation
5 {
6     /**
7      * @param args
8      */
9     public static void main(String[] args)
10    {
11        // print method
12        System.out.println("Hello Java");
13    }
14 }
15 }
16 }
17 }
```

The console output is:

```
<terminated> GEM01Calculation [Java Application] /Library/Java/JavaVirtualMachines/jdk-22.jdk/Contents/Home/bin/java (Aug 11, 2025, 6:02:49 PM - 6:02:50 PM) [pid:
Hello Java
```

Edit, Run and Compile



When you click Run (▶) in Eclipse, two things happen:

1. **Save & Compile** – Eclipse automatically saves your changes and compiles your .java source files into .class files (Java bytecode).
2. **Run on the JVM** – The compiled .class files are then executed by the Java Virtual Machine (JVM).

Compilation checks your code for syntax errors and turns it into a format the JVM can understand. If there are errors, Eclipse won't run the program until they're fixed.

Warnings don't stop compilation, but they should still be fixed to avoid future problems.

Editing Code:

- You can change your code anytime in the editor.
 - When you run the program again, Eclipse automatically recompiles the changed files.
 - You don't have to manually compile in most cases—Eclipse handles it for you.
- 

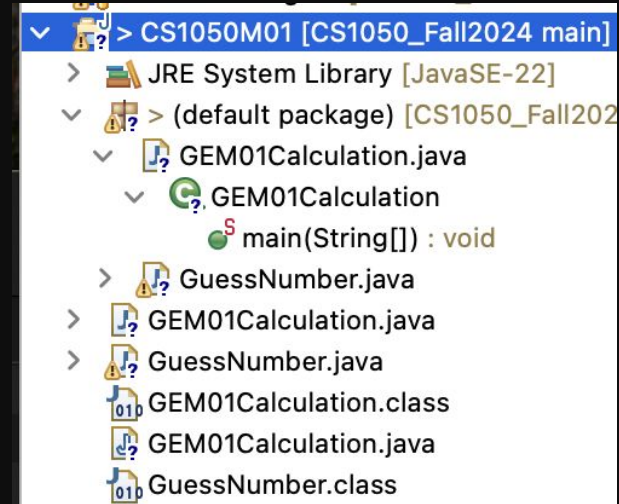
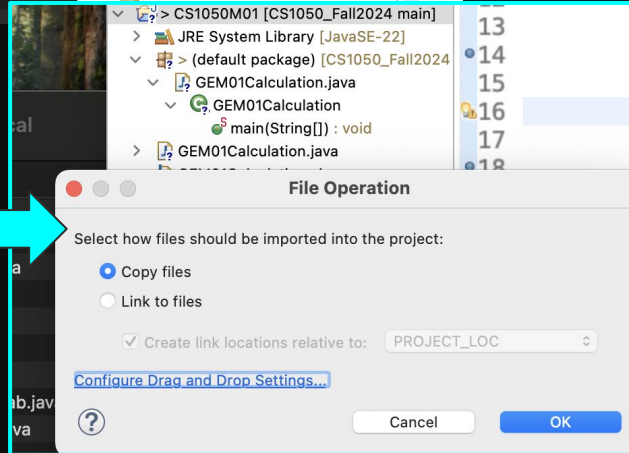
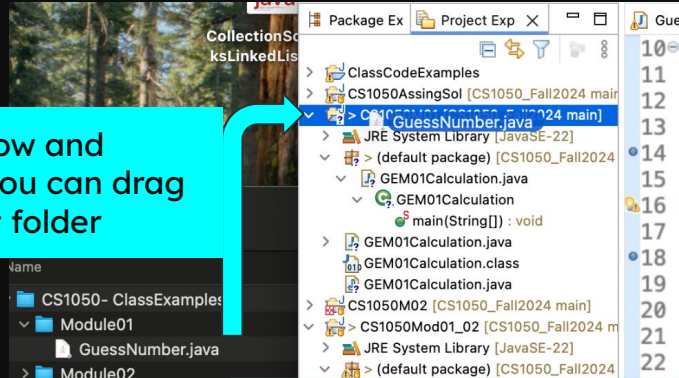
Add Existing Java File with Drag and Drop

You can add an existing Java file to a current project.

1. Make folder window and Eclipse window so you can drag java file into project folder

2. Click copy files so it will put the file in your repo workspace folder for versioning.

You should see the .java and .class file in the project. If you go to your github desktop you can commit and push a new version.



Errors

Error

Error Location - hover over for information on how to fix.

Problem

Problems Tab

Shows errors from all files.

Error Information

Description

File name

Line Number

Console Tab

Program won't run and displays error in Console window

The screenshot displays an IDE with a Java file named `GuessNumber.java`. The code contains a syntax error on line 18: `public static void main(String[] args) {`. The IDE highlights the error with a red squiggly line and a tooltip that says "Syntax error, insert ';' to complete BlockStatements".

The **Problems Tab** shows the error details in a table:

Description	Resource	Path	Location	Type
4 errors, 32 warnings, 0 others				
Errors (4 items)				
Syntax error, insert ";" to complete BlockStatements	GuessNumber.java	/CS1050Mod01_02	line 18	Java Problem

The **Console Tab** shows the error message when the program is run:

```
<terminated> GuessNumber (2) [Java Application] /Library/Java/JavaVirtualMachines/jdk-22.jdk/Contents/Home/bin/java (Aug 11, 2025, 6:09:22 PM - 6:09:22)
Exception in thread "main" java.lang.Error: Unresolved compilation problem:
    Syntax error, insert ";" to complete BlockStatements

    at GuessNumber.main(GuessNumber.java:18)
```

Warnings

Your program should not contain warnings. Why fix them?

- Possible Security Risks – Some warnings indicate unsafe code that could be exploited.
- Maintainability – Clean code is easier for you (and others) to read, debug, and improve.
- Professional Standards – In industry, code reviews often require zero warnings before approval.
- Prevent Future Bugs – Warnings can signal logic issues or outdated methods that might cause errors later.

The screenshot shows an IDE with a project named 'CS1050Mod01_02'. The 'GuessNumber.java' file is open, showing a resource leak warning: 'Resource leak: 'keyboardInput' is never closed'. The code snippet is as follows:

```
19  
20     int number = (int) (Math.random() * RANDOM_MULTIPLIER) + 1;  
21  
22     Scanner keyboardInput = new Scanner(System.in);  
23  
24     System.out.println("Guess a number between 0 and 100")  
25  
26     int guess = -1;  
27     while (guess != number)  
28     {  
29  
30         System.out.print("\nEnter your guess: ");  
31         guess = keyboardInput.nextInt();  
32     }  
33
```

The 'Problems' tab is open, showing a list of warnings. The table below represents the data shown in the 'Problems' tab:

Description	File name	Path	Location
Resource leak: 'input' is never closed	M02L06MathChars.java	/ClassCodeExamples	line 35
Resource leak: 'input' is never closed	project01iteration.java	/ClassCodeExamples	line 71
Resource leak: 'keyboardInput' is never closed	GuessNumber.java	/CS1050Mod01_02	line 22

warning

Warning Location - hover over
for information on how to fix.

Problems Tab

Problem

Warning Information

Shows warnings from all files

Description

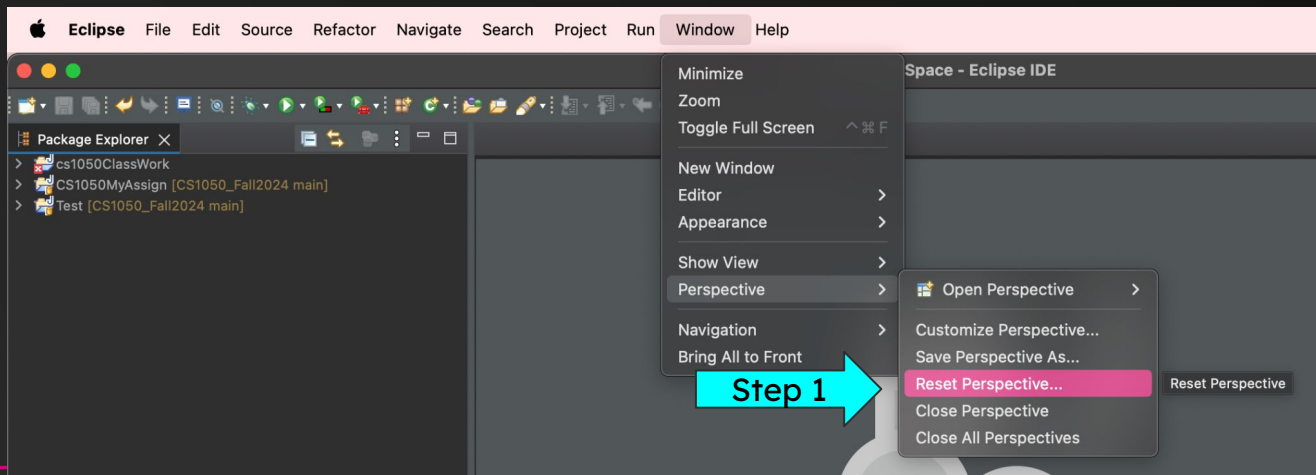
File name

Line Number

Perspective and Views

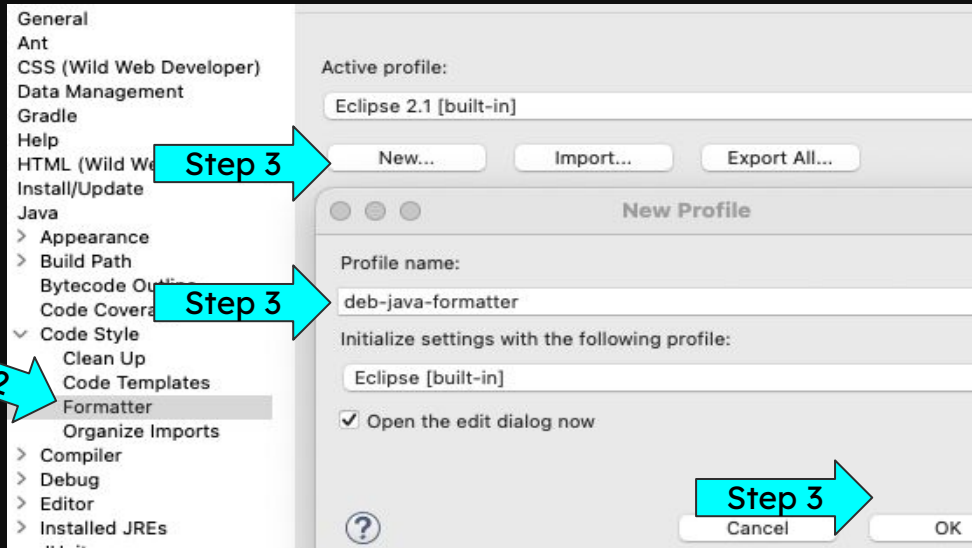
A “perspective” is just a layout of views and controls what you see in certain menus and toolbar. You will be using the Java and Debug Perspectives

1. If your “perspective” gets messed up Click window > perspective > reset perspective
 - a. If you accidentally close a panel and want to bring it back
 - b. If important windows like the console or package explorer disappear

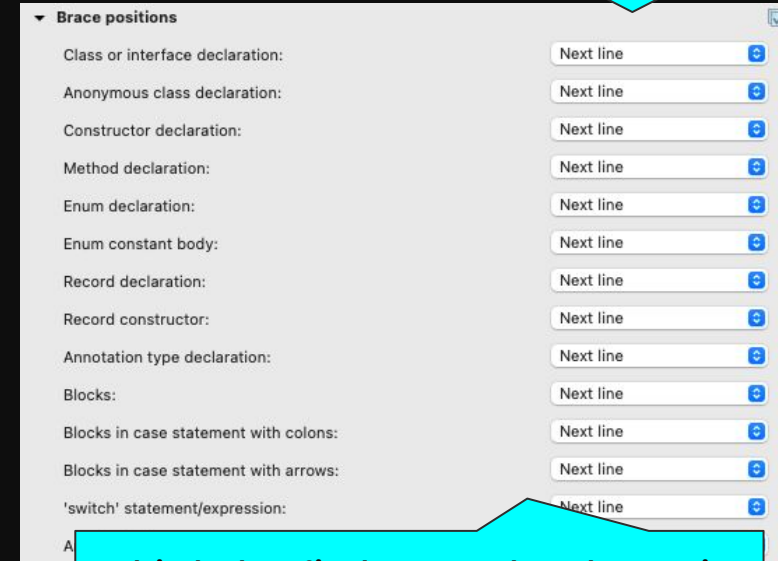


Formatter Profile Set Up

1. Go to settings (or might be called preferences)
 - a. Windows OS Go to menus Window -> preferences
 - b. Mac OS go to menus Eclipse -> settings
2. Then navigate to Java -> Code Style -> Format
3. Click new, Give your profile a name and click ok



4. Go to the Braces section, for all categories set Position to Next line.
5. Click ok to save profile.
6. Click apply and save



This helps find error when brace is missing or incorrectly placed.

Auto-Format on Save

Save Actions on, running code will always auto-format since Eclipse saves before compiling.

1. Go to
 - a. Mac: Eclipse → Settings → Java → Editor → Save Actions
 - b. Windows: Window → Preferences → Java → Editor → Save Actions
2. Select these save actions



Step 1

Save Actions

Step 2

- ☒ Perform the selected actions on save
- ☒ Format source code
 - ☒ Format all lines
 - ☐ Format edited lines

Configure the formatter settings on the [Formatter](#) page.
- ☒ Organize imports
 - Configure the organize imports settings on the [Organize Imports](#) page.
- ☐ Additional actions
 - Convert control statement bodies to block
 - Add missing '@Override' annotations
 - Add missing '@Override' annotations to implementations of interface methods
 - Add missing '@Deprecated' annotations
 - Add necessary casts

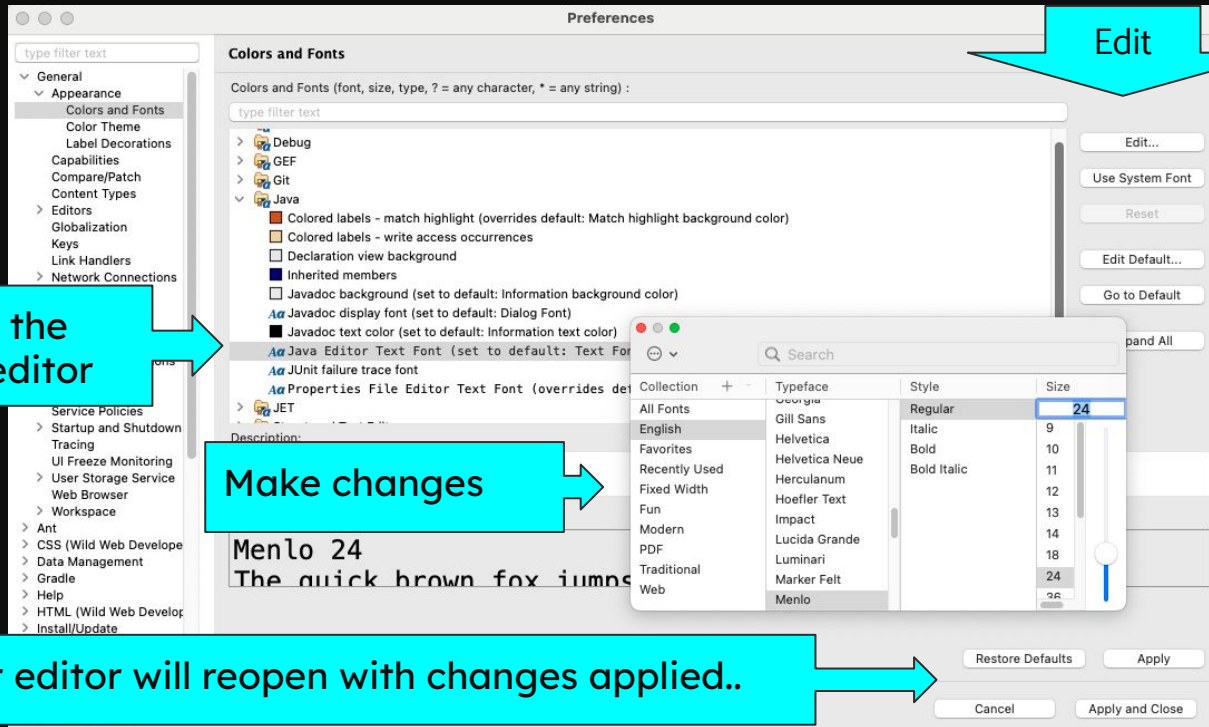
Type a messy block of code, then click Run — you should see the the correct format.

Change Themes, Colors Fonts

[6 Best Fonts for Coding to Keep Your Eyes from Eyestrain | by plabs.id](#)

If you want change the defaults go to Settings/Preferences general-> Appearance ->

- Colors and Fonts:
- Color Theme



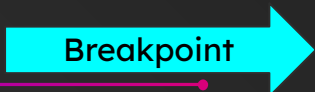
Debug - Set Breakpoints

Set Breakpoints to step through each statement and see what is stored in each variables memory

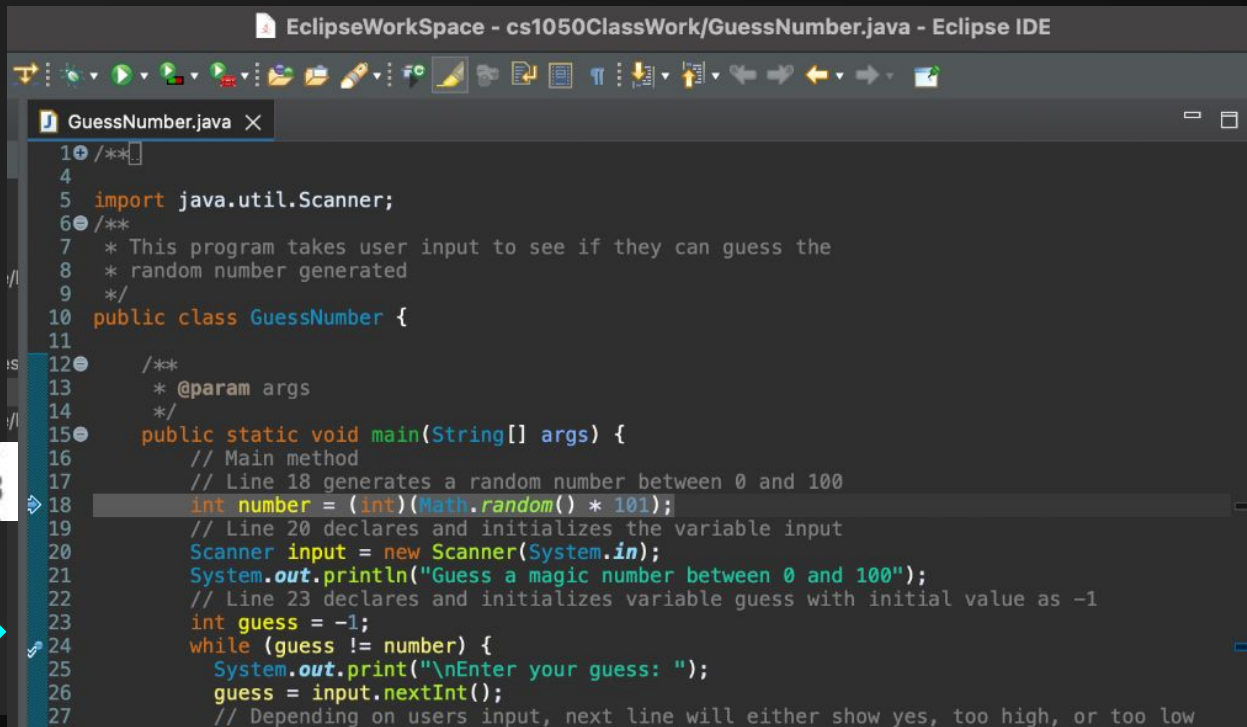
- To set a breakpoint, double click directly to the left of the line numbers
- A blue circle will appear next to the line, indicating that a breakpoint has been set
- To get rid of the breakpoint, double-click on it and the blue dot will disappear



Breakpoint



Breakpoint



```
1 /**
4
5 import java.util.Scanner;
6 /**
7  * This program takes user input to see if they can guess the
8  * random number generated
9  */
10 public class GuessNumber {
11
12     /**
13     * @param args
14     */
15     public static void main(String[] args) {
16         // Main method
17         // Line 18 generates a random number between 0 and 100
18         int number = (int)(Math.random() * 101);
19         // Line 20 declares and initializes the variable input
20         Scanner input = new Scanner(System.in);
21         System.out.println("Guess a magic number between 0 and 100");
22         // Line 23 declares and initializes variable guess with initial value as -1
23         int guess = -1;
24         while (guess != number) {
25             System.out.print("\nEnter your guess: ");
26             guess = input.nextInt();
27             // Depending on users input, next line will either show yes, too high, or too low
```

Debug: Start Debugging Session

To start a debugging session click the debug icon (a green bug) in the toolbar before the run button. You should see the variables window (Debug Perspective). If you don't this could be because

- You didn't create your code in a project
- The java code has syntax errors

Debug window

Variables window

Run Window Help

- ▶ Resume
- ⏸ Suspend
- ⏹ Terminate

Breakpoint Types

- Toggle Breakpoint
- Toggle Lambda Entry Breakpoint
- Toggle Tracepoint
- Toggle Line Breakpoint
- Toggle Watchpoint
- Toggle Method Breakpoint
- Skip All Breakpoints

GuessNumber.java

```
10 /**
11  *
12  * @param args
13  */
14 public static void main(String[] args) {
15     // Main method
16     // Line 18 generates a random number between 0 and 100
17     int number = (int)(Math.random() * 101);
18     // Line 20 declares and initializes the variable input
19     Scanner input = new Scanner(System.in);
20     System.out.println("Guess a magic number between 0 and 100");
21     // Line 23 declares and initializes variable guess with initial value as -1
22     int guess = -1;
23     while (guess != number) {
24         System.out.print("\nEnter your guess: ");
25     }
```

Variables

Name	Value
no method return value	
args	String[] (id=19)

Console

GuessNumber [Java Application] /Library/Java/JavaVirtualMachines/jdk-22.jdk/Contents/Home/bin/java (Mar 6, 2025, 9:30:07 PM elapsed: 0:00:47) [pid: 57759]

Do not select Skip All Breakpoints

If the program didn't stop you may not have set breakpoints or it is skipping breakpoints. Make sure Skip All Breakpoints is not turned on in Run menu.

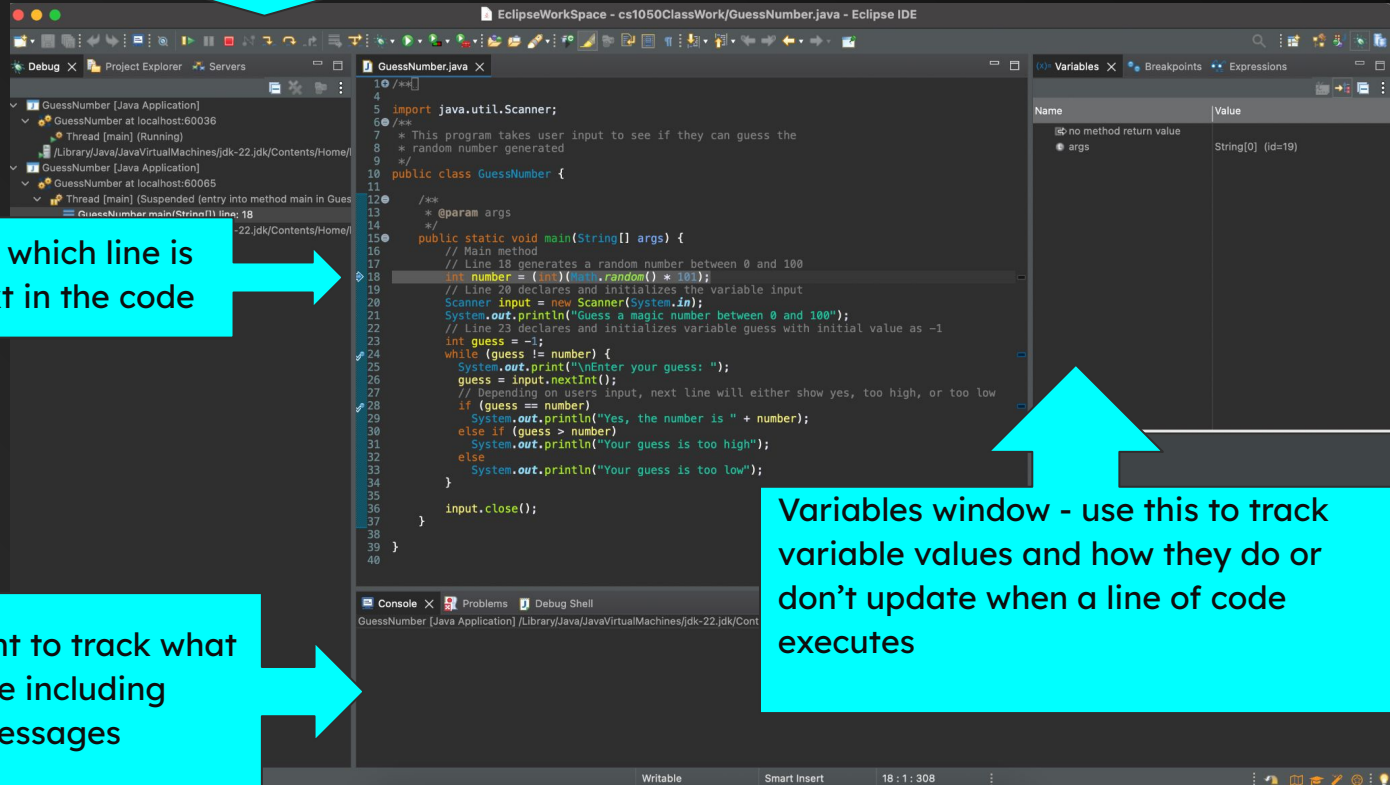
Debug: Step Through Code

Use the “step over” button to control the flow of the program and move through each line manually.

Menus



The pointer tells us which line is to be executed next in the code

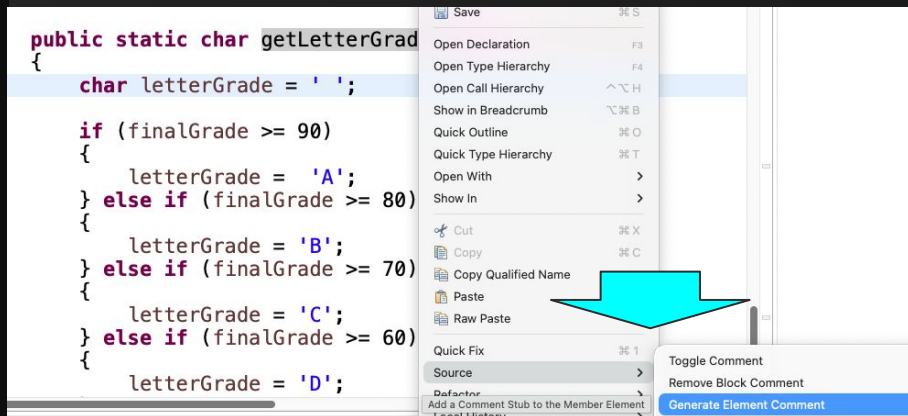


Variables window - use this to track variable values and how they do or don't update when a line of code executes

Console output - it's important to track what is being printed to the console including print statements and error messages

Generate Comments

Right Click inside a method and select Source -> Generate Element comment



/**

element comment

* @param

finalGrade

* @return

*/

public static char

getLetterGrade(double
finalGrade)

{

char

letterGrade = ' ';

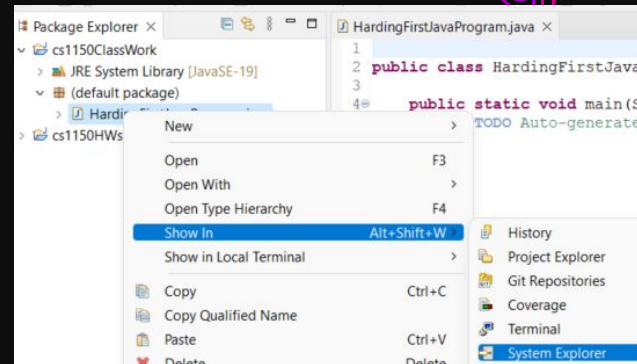
Keyboard Shortcuts

Action	Mac	Windows
Run Program	⌘ + F11	Ctrl + F11
Debug Program	F11	F11
Format Code	⌘ + ⇧ + F	Ctrl + Shift + F
Organize Imports	⌘ + ⇧ + O	Ctrl + Shift + O
Search / Find	⌘ + F	Ctrl + F
Find in Project	⌘ + H	Ctrl + H
Open Type (Class)	⌘ + ⇧ + T	Ctrl + Shift + T
Open Resource (File)	⌘ + ⇧ + R	Ctrl + Shift + R
Rename Variable/Method/Class	Alt + ⌘ + R	Alt + Shift + R
Switch Tabs	⌘ + Option + → / ←	Ctrl + PageDown / PageUp
Close Tab	⌘ + W	Ctrl + W
Undo	⌘ + Z	Ctrl + Z
Redo	⌘ + ⇧ + Z	Ctrl + Y

More Support

You can navigate to the file in your system explorer

- right click the file then go to Show In -> System Explorer



More Eclipse Resources

- [Java Debugging with Eclipse - Tutorial](#)
- [Eclipse Help](#)

“ How does a Java dev
cut his hair?

Eclipse it ”

@nhazelton

